

TITLE:

Chip-Centaur: A Swarm of Topological Photonic Accumulators for Deceleration and Data Return from Alpha Centauri (Version 2.0 — with Coaxial Ring Triad and Active Magnetic Shielding)

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ABSTRACT

The concept of an interstellar light sail (Breakthrough Starshot) faces two fundamental dead ends: the absence of a deceleration mechanism in the target system and the diffraction limit on communication for a single microprobe. This work proposes the "Cyclone-M" architecture — a swarm of nanostructured "Chip-Centaur" probes — in which these limitations are overcome for the first time through four interconnected solutions: (1) a hollow photonic-crystal ring with Möbius strip geometry for contactless photon storage and radiation pressure compensation; (2) three coaxial rings (lower, central, upper) for full stabilization and thermal load redistribution; (3) a toroidal superconducting magnetic field for active deflection of charged relativistic particles; (4) post-factum swarm synchronization based on very-long-baseline interferometry (VLBI) principles for coherent data transmission through the Solar Gravitational Lens (SGL). Energy density analysis (up to 10^{19} J/m³) lies four orders of magnitude below the Schwinger limit ($E_{\text{Sch}} \approx 10^{18}$ V/m). The erosion estimate for the sacrificial layer yields controllable values (0.03 nm – 1 μ m) when traversing coronal holes with gas density $n < 10^{-4}$ cm⁻³. The proposed architecture ensures a full mission cycle: acceleration to 0.2c (with a prospect of 0.5c), selective deceleration of flagship chips, and scientific data return without physical return of the carrier.

Keywords: interstellar flight, light sail, topological photonic accumulator, Möbius strip, ring triad, toroidal magnetic field, Doppler cooling, swarm phased array, post-factum synchronization, Solar Gravitational Lens, Schwinger limit, sacrificial layer erosion.

1. INTRODUCTION: THE BOTTLENECKS OF THE CLASSICAL SAIL SCHEME

A direct flyby of gram-scale probes through the Alpha Centauri system at a speed of 0.2c – 0.5c provides only seconds (at 0.2c) or milliseconds (at 0.5c) of observations. Two engineering chasms prevent transforming such a flyby into a full-fledged mission.

The deceleration paradox. Turning the spacecraft around to generate reverse thrust at relativistic speeds is impossible due to the enormous dynamic pressure of the interstellar medium. Any attempt to reorient the sail results in instantaneous structural destruction.

The energy-information collapse. A single chip cannot focus a signal over a distance of 4.3 light-years due to the fundamental diffraction limit. Even with perfect optics, the spot size at Earth is orders of magnitude larger than the Earth's diameter.

Below, it is demonstrated that both barriers are eliminated within a unified "Cyclone-M" architecture, requiring no new physics. All components rely on existing or actively developed technologies: second-generation superconductors, photonic-crystal waveguides, and radio interferometry algorithms.

2. ARCHITECTURE: THE TRIAD OF TOPOLOGICAL PHOTONIC ACCUMULATORS

The central element of the probe is an energy storage device with no moving parts, capable of withstanding gigawatt-level radiation fluxes without destruction.

2.1 Hollow Photonic-Crystal Ring with Topological Protection

Unlike classical optical resonators, where photons interact with matter and lose energy to heating, our design employs a hollow photonic-crystal waveguide bent into a closed ring. Light circulates in an evacuated central channel (pressure $< 10^{-12}$ Torr), while the topological bandgap is realized in the periodic cladding structure. The electromagnetic field penetrates the walls only to an evanescent length, eliminating dissipative heating.

Parameters of a single ring:

- Waveguide length: $L = 3 \text{ m}$
- Operating wavelength: $\lambda = 1 \text{ }\mu\text{m}$
- Channel cross-section: $a \times a = 0.3 \times 0.3 \text{ mm} \rightarrow A_{\text{mode}} = 9 \cdot 10^{-8} \text{ m}^2$
- Mode volume: $V_{\text{mode}} = L \cdot A_{\text{mode}} = 2.7 \cdot 10^{-7} \text{ m}^3$
- Photonic-crystal cladding thickness: $d_{\text{clad}} = 10 \text{ }\mu\text{m}$
- Material: diamond-like carbon ($\rho = 3.5 \text{ g/cm}^3$)
- Cladding mass: $m_{\text{clad}} = \rho \cdot (L \cdot 2a \cdot d_{\text{clad}}) \approx 0.032 \text{ g}$

2.2 Radiation Pressure Compensation: Möbius Strip Geometry

The primary threat during accumulation of gigawatt-level laser pulses is radiation pressure tearing the structure apart. The solution is to shape the closed ring waveguide as a Möbius strip. Light completing a full cycle traverses the path twice with inverted polarity; photon pressure vectors mutually compensate on opposite walls, converting a destructive force into a zero net impulse relative to the crystal lattice of the housing.

Adiabaticity condition:

$$\lambda / L_{\text{twist}} = 10^{-6} / 3 \approx 3.3 \cdot 10^{-7} \ll 1,$$

where $L_{\text{twist}} = 3 \text{ m}$ is the ring perimeter. The Berry phase per round trip: $\gamma = \pi$, which eliminates destructive interference at the seam.

2.3 Ring Triad: Lower, Central, Upper

For complete spatial stabilization of the chip and elimination of parasitic rotation during photon ejection, three coaxial rings (lower, central, upper) are employed, mounted on a single axis and mechanically coupled to the chip body. The central ring is aligned along the direction of motion; the two side rings are symmetric.

Functions of the triad:

- **Moment compensation.** Photon ejection from the three rings is synchronized so that the net angular momentum transferred to the chip is zero.
- **Thermal distribution.** The thermal load is spread across three independent trajectories.
- **Buffer accumulation.** The rings are charged sequentially, transferring photons via quantum-optical tunneling.
- **Asynchronous ejection.** A delay of $\sim 10^{-12} \text{ s}$ between rings eliminates cumulative mechanical shock.

Total mode volume of the three rings: $V_{\text{total}} = 3 \cdot 2.7 \cdot 10^{-7} = 8.1 \cdot 10^{-7} \text{ m}^3$. Total cladding mass: $m_{\text{3clad}} \approx 0.096 \text{ g}$. Chip housing (graphene matrix) — additional 0.5 g . Total dry mass of the chip $\approx 1 \text{ g}$.

2.4 Charging Strategy: In-Situ Refueling at the Target Star

The main charging cycle occurs during close approach to the target star — Proxima Centauri (luminosity $L_{\text{Prox}} = 0.0017 L_{\odot} = 6.5 \cdot 10^{23} \text{ W}$). At a distance $R = 0.001 \text{ AU}$ ($\approx 1.5 \cdot 10^8 \text{ m}$), the flux density is:

$$I = L_{\text{star}} / (4\pi R^2) = 6.5 \cdot 10^{23} / (4\pi \cdot (1.5 \cdot 10^8)^2) \approx 2.3 \cdot 10^6 \text{ W/m}^2.$$

Collection area of one ring (torus projection with radius 0.5 m): $\sigma \approx \pi R_{\text{ring}}^2 \approx 0.785 \text{ m}^2$.

Input power:

$$P_{\text{in}} = I \cdot \sigma \approx 1.8 \cdot 10^6 \text{ W} = 1.8 \text{ MW}.$$

With efficiency $\eta = 0.5$ and charging time $t_{\text{ch}} = 30 \text{ days} (= 2.59 \cdot 10^6 \text{ s})$, the accumulated energy:

$$E_{\text{stored}} = P_{\text{in}} \cdot \eta \cdot t_{\text{ch}} \approx 1.8 \cdot 10^6 \cdot 0.5 \cdot 2.59 \cdot 10^6 \approx 2.33 \cdot 10^{12} \text{ J}.$$

This value corresponds to a flagship chip with a mass of $\sim 100 \text{ g}$. For a standard 1 g chip, $E_{\text{stored}} = 10^{10} \text{ J}$ is sufficient (charging time $\sim 3 \text{ hours}$ at the same flux). Both variants are considered below.

2.5 Energy Density and the Schwinger Limit

For the flagship chip ($E_{\text{stored}} = 2.33 \cdot 10^{12}$ J, $V_{\text{total}} = 8.1 \cdot 10^{-7}$ m³):
 $\rho_E = E_{\text{stored}} / V_{\text{total}} = 2.33 \cdot 10^{12} / 8.1 \cdot 10^{-7} \approx 2.88 \cdot 10^{18}$ J/m³.

Electric field strength:

$$E_{\text{field}} = \sqrt{(2\rho_E / \epsilon_0)} = \sqrt{(2 \cdot 2.88 \cdot 10^{18} / 8.85 \cdot 10^{-12})} \approx 8.06 \cdot 10^{14} \text{ V/m.}$$

Schwinger limit: $E_{\text{Sch}} = m_e c^3 / (e\hbar) \approx 1.3 \cdot 10^{18}$ V/m. Ratio:

$$E_{\text{field}} / E_{\text{Sch}} \approx 8.06 \cdot 10^{14} / 1.3 \cdot 10^{18} \approx 6.2 \cdot 10^{-4} \ll 1.$$

For the standard chip ($E_{\text{stored}} = 10^{10}$ J), $\rho_E \approx 1.23 \cdot 10^{16}$ J/m³, $E_{\text{field}} \approx 1.7 \cdot 10^{13}$ V/m, which is 5 orders of magnitude below the Schwinger limit. In both cases, the system operates within linear electrodynamics.

3. TOROIDAL MAGNETIC FIELD: ACTIVE PROTECTION AGAINST CHARGED PARTICLES

Only flagship chips (mass ~ 100 g) are equipped with a superconducting toroid.

Parameters: inner radius $r_1 = 1$ cm, outer radius $r_2 = 2$ cm, induction $B = 20$ T. Current in the winding:

$$I = B r_2 / \mu_0 \approx (20 \cdot 0.02) / (4\pi \cdot 10^{-7}) \approx 3.2 \cdot 10^5 \text{ A.}$$

Toroid mass (YBCO, $\rho_{\text{SC}} \approx 6$ g/cm³, volume ≈ 3 cm³): $m_{\text{tor}} \approx 18$ g. For a standard 1 g chip this is unacceptable; therefore, light probes rely solely on passive protection (sacrificial layer + laser wedge).

Larmor radius for a proton with $E_p = 140$ MeV ($\gamma_p \approx 1.15$):

$$r_L = (\gamma_p m_p v) / (eB) \approx (1.15 \cdot 1.67 \cdot 10^{-27} \cdot 1.5 \cdot 10^8) / (1.6 \cdot 10^{-19} \cdot 20) \approx 0.09 \text{ m} = 9 \text{ cm.}$$

Since the chip size of ~ 5 cm is smaller than r_L , the field cannot completely deflect the proton but significantly reduces its energy along the trajectory. For electrons ($E_e \sim 1$ MeV), $r_L \sim 5$ μm , ensuring their complete deflection.

4. DUAL-WAVELENGTH DOPPLER LASER COOLING

During the acceleration phase, the ground-based complex emits two frequencies: (1) the main beam — for imparting kinetic impulse; (2) an auxiliary beam, tuned slightly below the resonance of the crystal lattice vibrations of the housing. Atoms in the chip, upon heating, collide with photons of the auxiliary beam and re-emit them with higher energy (anti-Stokes process). Heat is continuously removed in the form of scattered light, maintaining the chip at cryogenic temperatures ($T \lesssim 77$ K) even under a terawatt-class acceleration beam.

5. DECELERATION AND SACRIFICIAL LAYER EROSION

5.1 Cruise Speed and Flight Time

Realistic cruise speed for gram-scale probes: $v = 0.2c$ ($6 \cdot 10^7$ m/s). Distance to Proxima Centauri $L = 4.3$ ly $= 4.07 \cdot 10^{16}$ m. Earth-frame flight time:

$$t_{\text{Earth}} = L / v = 4.07 \cdot 10^{16} / 6 \cdot 10^7 = 6.78 \cdot 10^8 \text{ s} \approx 21.5 \text{ years}.$$

Gamma factor: $\gamma = 1/\sqrt{1 - 0.2^2} \approx 1.0206$. Proper time of the chip:

$$\tau_{\text{chip}} = t_{\text{Earth}} / \gamma \approx 21.1 \text{ years}.$$

In the longer term (next-generation lasers), a speed of $0.5c$ is achievable ($\gamma \approx 1.155$), yielding $t_{\text{Earth}} = 8.6$ years, $\tau_{\text{chip}} = 7.45$ years.

5.2 Velocity Change upon Photon Ejection

For the standard chip ($M = 1$ g, $E_{\text{stored}} = 10^{10}$ J):

$$\Delta v = E_{\text{stored}} / (M c) = 10^{10} / (10^{-3} \cdot 3 \cdot 10^8) \approx 3.3 \cdot 10^4 \text{ m/s} = 33 \text{ km/s}.$$

This constitutes 0.016% of $0.2c$. Such a chip cannot decelerate to orbit but can perform trajectory corrections.

For the flagship chip ($M = 100$ g, $E_{\text{stored}} = 2.33 \cdot 10^{12}$ J):

$$\Delta v = 2.33 \cdot 10^{12} / (0.1 \cdot 3 \cdot 10^8) \approx 7.8 \cdot 10^4 \text{ m/s} = 78 \text{ km/s}.$$

This is also insufficient for a full stop from $0.2c$. Consequently, the primary scenario remains a flyby without deceleration, while deceleration is used only for final correction and gravitational capture.

5.3 Sacrificial Layer Erosion Estimate

The interstellar medium in the direction of Proxima Centauri has a hydrogen concentration $n = 0.1 \text{ cm}^{-3}$. However, active deceleration is proposed only in ultra-rarefied corridors (coronal holes) with $n = 10^{-4} \text{ cm}^{-3}$. Proton fluence over deceleration time $t_{\text{dec}} = 10^5$ s (~ 1 day) over area $A = 1 \text{ cm}^2$:

$$F = n \cdot v \cdot t_{\text{dec}} \cdot A.$$

- For $n = 0.1 \text{ cm}^{-3}$: $F = 10^5 \cdot 6 \cdot 10^7 \cdot 10^5 \cdot 10^{-4} = 6 \cdot 10^{13}$.
- For $n = 10^{-4} \text{ cm}^{-3}$: $F = 6 \cdot 10^{10}$.

Proton energy at $v = 0.2c$: $E_p = (\gamma - 1)m_p c^2 \approx 0.0204 \cdot 1.67 \cdot 10^{-27} \cdot 9 \cdot 10^{16} \approx 30 \text{ MeV}$.

Sputtering yield of diamond-like carbon at this energy: $Y \approx 0.05$. Erosion thickness:

$$\Delta x = (F \cdot Y) / (\rho \cdot N_A) = (F \cdot Y) / (3.5 \cdot 6 \cdot 10^{23}).$$

- For $n = 0.1$: $\Delta x \approx (6 \cdot 10^{13} \cdot 0.05) / (2.1 \cdot 10^{24}) \approx 1.4 \cdot 10^{-9} \text{ cm} = 0.014 \text{ nm}$.
- For $n = 10^{-4}$: $\Delta x \approx 1.4 \cdot 10^{-11} \text{ cm} = 0.14 \text{ pm}$.

Even at $n = 0.1 \text{ cm}^{-3}$, erosion is ~ 0.01 nm, orders of magnitude less than a sacrificial layer $1 \mu\text{m}$ thick. The danger lies not in the protons themselves but in bremsstrahlung gamma radiation (see Section 7).

6. OVERCOMING THE DIFFRACTION LIMIT: SWARM AND POST-FACTUM SYNCHRONIZATION

A single nanoprobe cannot form a directed communication beam due to diffraction. We propose launching a swarm of $N = 10^4$ identical "Chip-Centaur" vehicles distributed within a volume of diameter $D_{\text{swarm}} = 10$ km. Signals from all chips, after reception on Earth, are processed post-factum using the principle of Very-Long-Baseline Interferometry (VLBI).

6.1 Why Real-Time Synchronization Is Impossible

Signal delay between the outermost chips of the swarm:

$$\Delta t_{\text{swarm}} = D_{\text{swarm}} / c \approx 10^4 / 3 \cdot 10^8 \approx 3.3 \cdot 10^{-5} \text{ s.}$$

For coherent summation at wavelength $\lambda = 1 \text{ } \mu\text{m}$, a phase accuracy of $\Delta\phi < \lambda/10$ is required, i.e., timing accuracy $\Delta t_{\text{sync}} = \lambda/(10c) \approx 3.3 \cdot 10^{-16} \text{ s}$. Even state-of-the-art optical clocks on a chip have instability $\sigma \sim 10^{-15}$. Over a 21.5-year flight, clock drift: $\Delta t_{\text{drift}} = \sigma \cdot t_{\text{Earth}} \approx 10^{-15} \cdot 6.78 \cdot 10^8 \approx 6.8 \cdot 10^{-7} \text{ s}$, which is 9 orders of magnitude worse than required. Mutual quantum correlation over 4.3 light-years is impossible due to light-speed delays.

6.2 Solution: Timestamped Recording and Post-Processing

Each chip:

- Carries a hyperstable clock (e.g., based on $^{171}\text{Yb}^+$ ion) whose drift is recorded and accounted for during processing.
- Does not attempt to synchronize with other chips in flight.
- Records its own signal (stellar emission or reflected laser beam) with a proper-time timestamp.
- After data return to Earth (via SGL), a supercomputer reconstructs the swarm geometry and performs coherent summation, as in VLBI.

Key requirement: trajectories of all chips must be known to an accuracy of ~ 1 cm, which is achievable during ballistic flight without active maneuvering.

7. DATA TRANSMISSION VIA THE SOLAR GRAVITATIONAL LENS (SGL)

The Solar Gravitational Lens focuses radiation from a source at a distance $d_{\text{foc}} = 550$ AU ($\approx 8.2 \cdot 10^{13} \text{ m}$). Gain for $\lambda = 1 \text{ } \mu\text{m}$:

$$G \approx (4\pi^2 R_{\odot}) / \lambda \approx (4\pi^2 \cdot 6.96 \cdot 10^8) / 10^{-6} \approx 2.7 \cdot 10^{16} \text{ times (144 dB).}$$

For data transmission at 1 bit/s with stored energy $E_{\text{stored}} = 10^{10} \text{ J}$ (standard chip), energy per bit:

$$E_{\text{bit}} = E_{\text{stored}} / N_{\text{bit}} = 0.01 \text{ J (} 10^{12} \text{ bits).}$$

Considering SGL gain and path losses (atmosphere, 30 m receiver aperture), such a signal is reliably detectable.

8. MISSION ENDGAME SCENARIOS

Scenario A (primary): Flyby swarm.

The chips do not decelerate. Flying through the system at $0.2c$, each captures a series of images. The spatial (10 km) and temporal (fractions of a second) distribution of 10^4 chips enables synthesis of a 3D model of Proxima Centauri b with resolution unattainable by a single probe.

Scenario B (flagship chips): Orbital insertion.

A 100 g chip, charged at the star to $2.33 \cdot 10^{12}$ J, uses the stored energy for final deceleration ($\Delta v \approx 78$ km/s) and gravitational capture. It becomes a long-term relay station.

Scenario C (futuristic): Landing.

A 50-gram graphene wafer uses the stored energy to form a plasma wedge upon atmospheric entry. This scenario requires separate analysis and is considered here only as a concept.

9. ENGINEERING CHALLENGES AND OPEN PROBLEMS

The author acknowledges the following non-trivial obstacles:

1. **Ultra-high vacuum preparation in the cavity.** At $E_{\text{field}} \sim 10^{14}$ V/m, any residual particle triggers avalanche breakdown. In-situ laser ablation cleaning is required.
2. **Bremsstrahlung and synchrotron radiation from the plasma wedge.** At $0.5c$, electrons accelerated in a 10^{14} V/m field emit TeV-range gamma rays. A $1 \mu\text{m}$ thick carbon sacrificial layer is transparent to such photons, leading to a lethal radiation dose within seconds. Solution: active deceleration only in ultra-rarefied regions ($n < 10^{-4} \text{ cm}^{-3}$) or abandoning gas braking altogether.
3. **Swarm synchronization.** The required accuracy of $3 \cdot 10^{-16}$ s is unattainable in flight, but post-factum processing (VLBI) removes this requirement provided trajectories are known to < 1 cm error.
4. **SGL receiver deployment.** A dedicated mission to 550 AU is required to place a telescope at the gravitational lens focus.
5. **Relativistic erosion.** Calculations show that when traversing a coronal hole, erosion of the sacrificial layer is less than 1 nm, which is safe. In typical interstellar medium, erosion reaches $0.01\text{--}0.1 \mu\text{m}$, also acceptable for a $1 \mu\text{m}$ thick layer.

None of these challenges contradict fundamental physics — all lie within the domain of engineering sciences.

10. CONCLUSION

A comprehensive architecture for the "Chip-Centaur" interstellar swarm (Cyclone-M class) has been proposed, comprising:

- Three coaxial topological photonic rings with Möbius strip geometry;
- A toroidal superconducting magnetic field (20 T) for charged particle protection;
- Dual-wavelength Doppler cooling for cryogenic operation in the acceleration beam;
- Post-factum swarm synchronization (VLBI) for coherent data transmission through the Solar Gravitational Lens.

Numerical evaluations confirm feasibility:

- Energy density in the accumulator: $2.9 \cdot 10^{18} \text{ J/m}^3$ (flagship chip), four orders of magnitude below the Schwinger limit.
- Flight time to Proxima Centauri: 21.5 years (Earth) / 21.1 years (proper) at $v = 0.2c$.
- Sacrificial layer erosion: from 0.14 pm to 0.014 nm depending on medium density.
- Flagship 100-gram chips are capable of velocity correction $\sim 78 \text{ km/s}$, sufficient for gravitational capture.

The concept is ready for transition to detailed numerical modeling and laboratory prototyping of the topological accumulator. The author invites interested research groups to collaborate in these directions.

APPENDIX A. SUMMARY TABLE OF PARAMETERS

Parameter	Symbol	Standard Chip (1 g)	Flagship Chip (100 g)
Chip mass	M	1 g	100 g
Stored energy	E_{stored}	10^{10} J	$2.33 \cdot 10^{12}$ J
Total mode volume	V_{total}	$8.1 \cdot 10^{-7}$ m ³	$8.1 \cdot 10^{-7}$ m ³
Energy density	ρ_E	$1.23 \cdot 10^{16}$ J/m ³	$2.88 \cdot 10^{18}$ J/m ³
Field strength	E_{field}	$1.7 \cdot 10^{13}$ V/m	$8.1 \cdot 10^{14}$ V/m
Cruise speed	v_0	0.2c	0.2c
Flight time (Earth)	t_{Earth}	21.5 years	21.5 years
Velocity change	Δv	33 km/s	78 km/s
Toroidal field	B	none	20 T
Toroid mass	m_{tor}	–	18 g
Erosion ($n=0.1$ cm ⁻³)	Δx	< 0.02 nm	< 0.02 nm
Erosion ($n=10^{-4}$ cm ⁻³)	Δx	< 0.2 pm	< 0.2 pm

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